Sahil Katiyar

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PROFESSIONAL SUMMARY

B.Sc. Data Science graduate building and evaluating production-grade AI systems — multi-agent RAG pipelines, MLOps on AWS, and empirical benchmarking of LLM agent techniques (RAGAS, LLM-as-judge). Comfortable across the stack: designing **agent architectures (LangGraph), deploying them (Docker, Kubernetes, CI/CD)**, and proving they work with data, not assumptions. Seeking an AI Engineering internship to bring that build-then-measure approach to real-world products.

EDUCATION

Degree	Board	Percentage/CGPA	Year
Bachelors of Data Science and Analytics	Mumbai University (R.A. Podar college of Commerce and Economics)	10.00/10.00	2026

SKILLS

Languages: Python, SQL

ML/DL: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLP

LLMs & Agents: LangChain, LangGraph, LangSmith, RAG, Prompt Engineering, Tool-Calling, Multi-Agent Orchestration, MCP, Context Engineering, Hugging Face, Transformers, Unsloth, LoRA/QLoRA Fine-tuning, Embeddings.

DevOps & Cloud: Docker, Kubernetes, AWS (EC2, S3, EKS, IAM), FastAPI, Git, GitHub Actions, CI/CD, MLflow, MLOps

Data & Retrieval: Vector Databases (Chroma), Neo4j, | Evaluation: RAGAS, LLM-as-a-Judge

PROJECTS

Boss–Employee Agentic RAG System | Github

- Built a multi-agent AI system simulating an office workflow, where a **boss agent** dynamically assigns tasks to **employee agents (Data Analysis, Research)** powered by Groq, Gemini api.
- Designed an **Agentic Retrieval-Augmented Generation (RAG)** pipeline that adaptively **routes queries across SQL, Chroma, and Neo4j** based on the nature of the question, using **vector search with reranking to improve retrieval accuracy** on the Chroma-based retrieval path. Enabled multi-step reasoning and planning using agent-based workflows to improve task execution and contextual response quality.
- Integrated **observability using LangSmith** to monitor errors, latency, and workflow execution, achieving ~5–15 seconds response time depending on query complexity.

CFPB Complaint Intelligence System — End-to-end MLOps pipeline | Github

- Built and deployed dual NLP models (PyTorch MLP, Scikit-learn MLP) for complaint product classification (**86% F1**) and urgency detection (**85% F1, 99% precision**).
- Designed an NLP preprocessing and feature engineering pipeline with **MLflow-based experiment tracking and model versioning. Automated containerized deployment to AWS EKS using Docker, Kubernetes, and a full CI/CD pipeline via GitHub Actions.**
- Troubleshot and resolved production infrastructure issues (node sizing, memory limits, AWS account constraints) to ensure reliable model serving.

Context Engineering Agent – Benchmarking Retrieval Compression & Memory for LLM Agents | Github

- Built a LangGraph agent to test whether **context engineering** (filter → dedupe → re-rank via Cohere → compress) improves performance, with **short-term (checkpointer)** and **long-term memory (Mem0)**.
- **Benchmarked 7 pipeline** variants using **LLM-as-judge and RAGAS metrics, cutting token usage ~44%** while holding answer quality nearly constant (4.39 → 4.28) vs. baseline. Validated the relevance filter under a real failure case, correctly dropping irrelevant search results before they reached the prompt.

AI-Driven Demand Forecasting & Supply Chain Optimization for Power Grid Infrastructure | Github

- Developed an AI-driven supply chain platform for power grid infrastructure, combining a **multi-output ML model with a multi-agent GenAI system to automate procurement planning, logistics, and risk analysis.**
- Built an ExtraTreeRegressor model predicting multiple raw materials (steel, conductors, concrete) simultaneously, and designed a **LangGraph-orchestrated multi-agent pipeline** — Price (Alpha Vantage, ExchangeRate-API), Logistics (Geoapify), and Risk (Google Gemini, SerpApi) agents.
- Integrated a **conversational AI orchestrator powered by Llama-3-70B** (via Groq) for real-time plan Q&A, and deployed via Docker for production-ready execution, achieving ~5–20 seconds response time.

Langgraph_AI_notes_workflow | Github

Self-correcting agents, **human-in-the-loop**, multi-LLM orchestration (Gemini/Mistral), stateful checkpointing.

CO-CURRICULAR/CERTIFICATES

- **Google's Genai Exchange Hackathon**: worked with **Google Cloud** and made my first gen AI project - RAG.
- Led CodeCrux, the college's first inter-university coding competition, managing the organizing team.